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Vision -Based Smart Garbage Segregation System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

ITA0515- Computer Vision

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**VISION-BASED SMART GARBAGE SEGREGATION SYSTEM**” has been carried out by **PAVAN JANA (192472313)** and **V. RAGHAVENDRA (192421354)** under the supervision of **DR. Carmel Mary Belinda M** and is submitted in partial fulfilment of the requirements for the current semester of the **B. Tech Computer Science Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **PAVAN JANA and V. RAGHAVENDRA** of the **B.Tech Computer Science Engineering**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**VISION-BASED SMART GARBAGE SEGREGATION SYSTEM**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 5.09.26

Signature of the Students with Names

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
PAVAN JANA	Image upload & preprocessing	Upload, validation, resizing, preprocessing	Image format & output testing	Ch. 1, 3, 4	50 %
V.RAGHAVENDRA	Architecture & system support	Integration & preprocessing support	Functional testing & analysis	Ch. 2, 3, 5	50 %
Total					100%

ABSTRACT

Improper segregation of solid waste is a major environmental and operational problem because mixed garbage reduces the efficiency of recycling and increases the amount of waste sent to landfills. Vision-based garbage segregation systems can assist in identifying different waste categories from images. However, the quality and consistency of the input image have a direct influence on the performance of subsequent image-classification algorithms. Therefore, an effective image uploading and preprocessing module is required as the initial stage of a vision-based waste segregation system.

This project develops an Image Uploading and Preprocessing Module for a Vision-Based Smart Garbage Segregation System. The module provides a user interface through which a garbage image can be uploaded and validated before processing. The uploaded image undergoes preprocessing operations including image format verification, resizing, noise reduction, color-space processing and pixel normalization. These operations convert images with different resolutions and visual characteristics into a standardized representation suitable for further computer-vision and machine-learning processing.

The methodology consists of image acquisition, file validation, image decoding, resizing, preprocessing, normalization and generation of the processed image. The module is designed using Python-based image-processing techniques and can be integrated with a subsequent garbage-classification model. Testing is performed using different image formats, resolutions and image conditions to verify successful uploading and preprocessing.

The developed module provides a structured and consistent input to the later stages of the garbage segregation system. It reduces variations caused by image dimensions, format and pixel representation and establishes a reliable interface between image acquisition and garbage classification. The principal contribution of this module is therefore the development of a standardized image-processing pipeline that can serve as the foundation for an intelligent vision-based garbage segregation system.

Keywords: Computer Vision, Garbage Segregation, Image Processing, Image Preprocessing, Waste Classification, Machine Learning

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	-
CV	Computer Vision	-
FPS	Frames Per Second	frames/s

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Solid waste contains different materials such as plastic, paper, glass, metal and organic waste. When these materials are mixed, recycling and waste-treatment processes become more difficult. Waste segregation at an early stage can improve recycling efficiency and reduce unnecessary landfill disposal.

Traditional waste segregation depends heavily on manual sorting. Manual sorting is time-consuming, labour-intensive and can expose workers to unpleasant or potentially hazardous waste. Computer vision provides an alternative approach in which images of waste materials can be captured and processed automatically.

1.2 Need for the Project

The need for the project arises from the difficulty of efficiently processing mixed waste and identifying recyclable or non-recyclable materials.

- The major reasons for developing the system are:
- Manual segregation requires significant human effort.
- Waste materials may be mixed together.
- Manual identification can be inconsistent.
- Images obtained from different sources have different resolutions.
- Machine-learning models generally require standardized input.

1.3 Problem Statement

Waste images acquired from different sources may vary considerably in resolution, dimensions, colour representation and illumination. If these variations are directly supplied to a machine-learning model, they can increase input inconsistency and affect subsequent classification performance.

Therefore, the engineering problem is to design and implement a reliable image uploading and preprocessing module capable of accepting valid waste images, validating the input and transforming them into standardized image representations suitable for subsequent computer-vision-based garbage classification.

The problem involves several interacting factors:

- Image format compatibility.
- Different image dimensions.
- Different resolutions.
- Pixel-value variation.

1.4 Project Aim

The aim of this project is to develop an image uploading and preprocessing module that prepares garbage images in a standardized format for subsequent computer-vision-based waste classification and segregation.

1.5 Project Objectives

- To design an image-uploading interface for waste images.
- To validate uploaded image files before processing.
- To resize input images to a standardized dimension.
- To convert images into a suitable colour representation.
- To normalize image data for consistent processing.
- To generate a standardized preprocessed image.

1.6 Scope

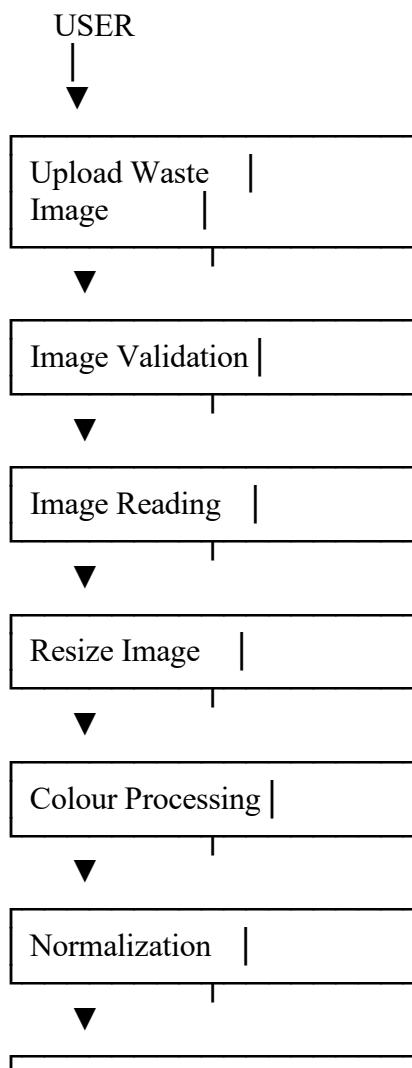
Clearly state:

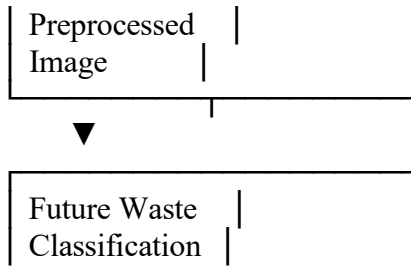
- **Included:**
 - Image uploading.
 - Image-format validation.
 - Image reading.
 - Image resizing.
- **Excluded:**
 - The present implementation does not include:
 - Automatic waste classification.
 - Robotic waste picking.
 - Conveyor-belt segregation.

- Physical waste bins with actuators.
- **Assumptions:**
 - he uploaded file contains a recognizable waste object.
 - The image is readable by the software.
 - The uploaded image uses a supported format.
 - The computer has sufficient processing resources.
- **Boundary conditions:**
 - The module operates on digital image files supplied by the user. Its performance depends on image quality, file format, resolution and the visibility of the waste object.

1.7 Expected Engineering Outcome

Define the intended:





CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Briefly describe how relevant literature, standards, patents, products or existing technologies were identified. Literature was reviewed using research papers and technical resources related to computer vision, image preprocessing, machine-learning-based waste classification and smart waste-management systems.

The review focused on:

- Waste classification.
- Image-based waste recognition.
- Image preprocessing.
- CNN-based classification.
- Dataset preparation.

2.2 Review of Existing Work

Image-Based Waste Classification:

Vision-based waste classification systems use images of waste objects as input and apply machine-learning or deep-learning techniques to identify the material category. Common categories include cardboard, glass, metal, paper, plastic and organic waste.

A major requirement of such systems is consistent image preprocessing. Classification models generally expect images with fixed dimensions and a consistent numerical representation.

Convolutional Neural Networks:

CNNs are widely used for image classification because they can automatically learn spatial features from images. A CNN-based waste classifier can receive the preprocessed image and predict the most likely waste category.

Image Resizing:

Images captured or uploaded by users can have different dimensions. Resizing converts these images to a fixed resolution required by the subsequent model.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual waste identification	Simple and requires little technology	Time-consuming and inconsistent	Baseline
Basic image resizing	Easy and computationally inexpensive	Does not address all input variations	Included

2.4 Research / Engineering Gap

Many vision-based waste-management systems focus mainly on the final classification model. However, reliable classification also depends on the quality and consistency of the input images. The engineering gap addressed by this project is the development of a structured image-uploading and preprocessing pipeline that prepares user-provided waste images before classification. The module establishes a clear interface between raw user input and a future classification model.

2.5 Proposed Contribution

The proposed work contributes an independent preprocessing module consisting of:

Upload → Validation → Image Reading → Resizing → Colour Processing → Normalization → Processed Output

This modular structure allows a future machine-learning classifier to be integrated without redesigning the image-input stage.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Developer	Modular preprocessing pipeline

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Upload image	Functional	Supported image accepted

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Software quality considerations	Usability, reliability and performance considered

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Input format	JPG/PNG and other supported formats	Format	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Direct Image Upload Without Preprocessing

The uploaded image would be passed directly to the next stage. This is simple but creates inconsistent image dimensions and pixel representations.

Alternative 2: Complete Preprocessing Pipeline

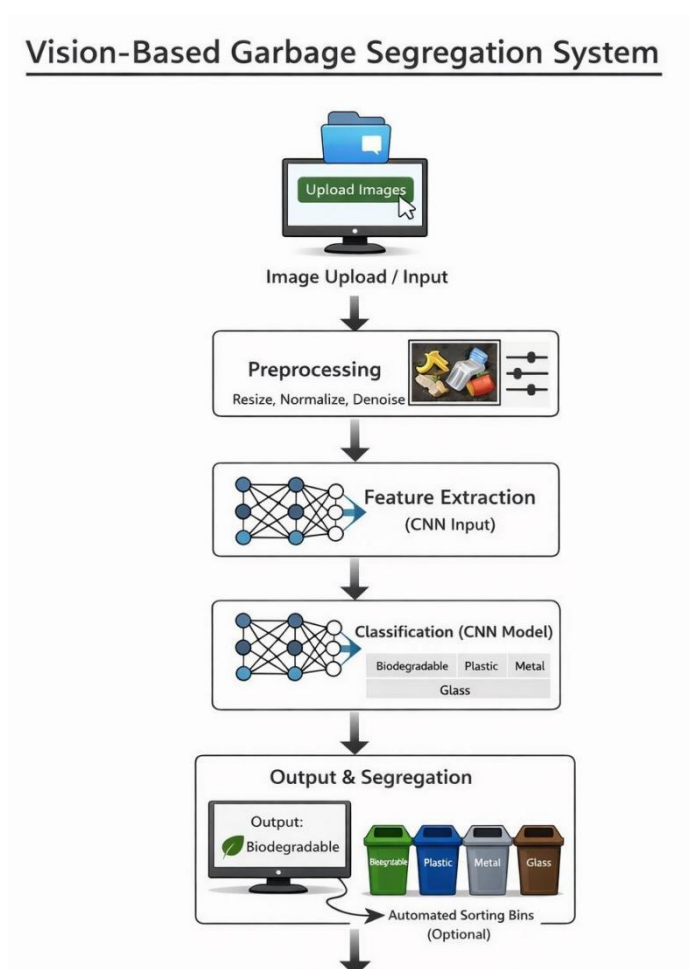
The image is validated, resized, converted into the required colour representation and normalized before being supplied to the next stage.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	5	5	4
Cost	20%	5	5	5
Safety	20%	2	4	5
Sustainability	15%	5	5	5
Reliability	20%	2	4	5

3.5 Selected Design & Detailed Design

The final design uses a modular image uploading and preprocessing architecture. The uploaded garbage image is first validated and then processed through resizing, colour conversion and pixel normalization. This design was selected because it provides a consistent image format for the subsequent waste-classification stage, is simple to implement, requires low computational resources and allows each processing stage to be tested independently.

Insert architecture diagram / schematic / flowchart here.



Systems approach:

The upload interface provides the raw image to the preprocessing module. The validation stage checks whether the uploaded file can be processed. The preprocessing stage then performs resizing, colour conversion and normalization according to the requirements of the future classification model. The final processed image becomes the standardized interface between image acquisition and waste classification.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Image input	Camera only	Automatic acquisition	Requires camera access	Upload selected for flexible testing

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Software-based image processing requires no physical sorting equipment during the current implementation	Only a computer and software libraries are required	A software-based approach was selected to reduce material and hardware requirements
Ethics	Uploaded images may unintentionally contain personal information	Images are used for waste-image processing	Avoid unnecessary storage and sharing of uploaded images
Health & Safety	Current implementation does not involve direct physical waste handling or moving machinery	Processing is performed digitally	Low physical safety risk; mechanical segregation is outside the current scope
Security	Unsupported or corrupted files may create application and data-security risks	Image files are validated before processing	File validation and controlled image handling are included in the design

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Corrupted image	Medium	Medium	Medium	Validate image before processing	Low

CHAPTER 4

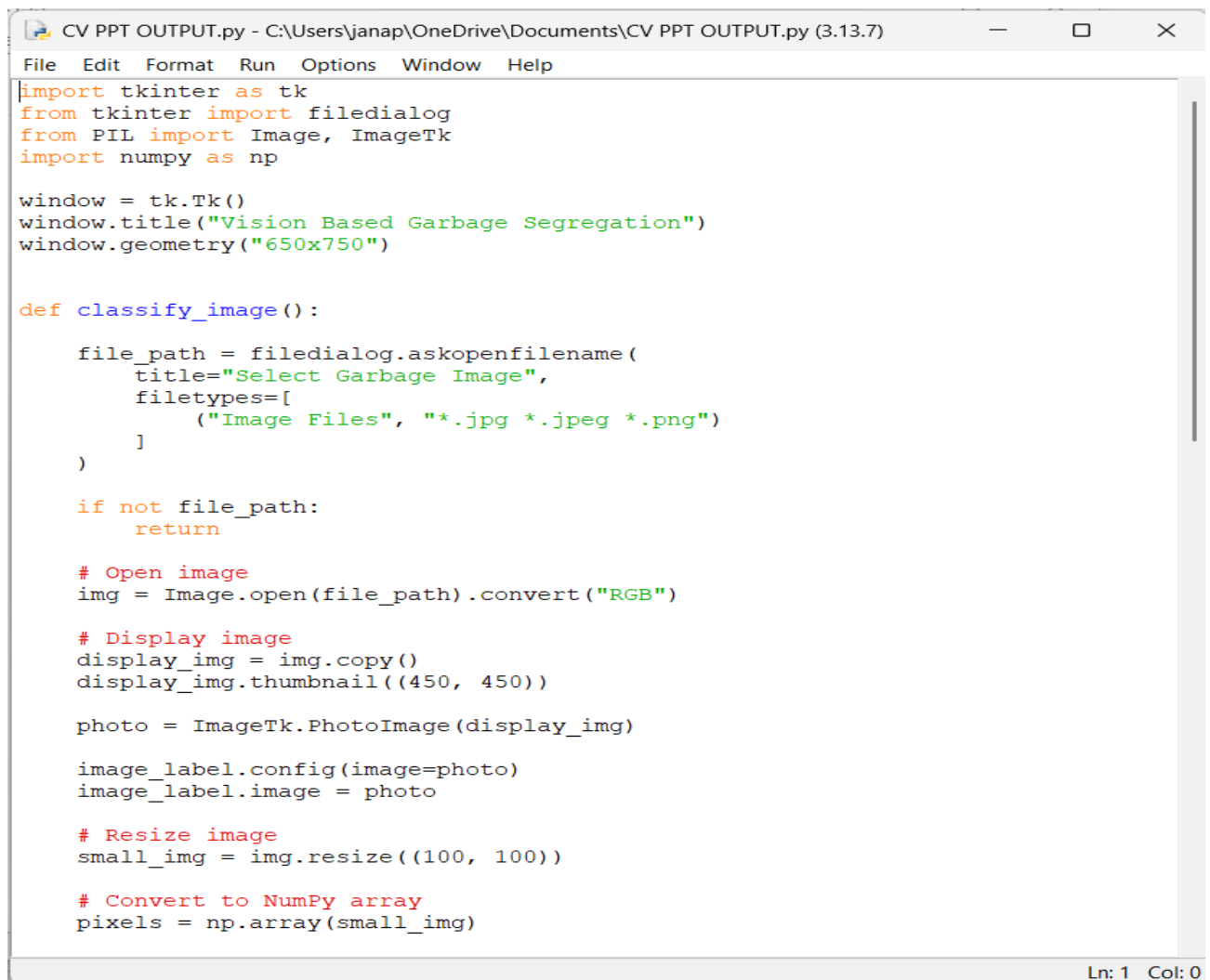
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The module was developed using an incremental approach. First, the image-uploading mechanism was established, followed by input validation and image loading. The preprocessing stages were then implemented to standardize the image dimensions and pixel representation. Each stage was tested individually before integrating the complete preprocessing pipeline.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Main programming language	Development



```
CV PPT OUTPUT.py - C:\Users\janap\OneDrive\Documents\CV PPT OUTPUT.py (3.13.7)
File Edit Format Run Options Window Help
import tkinter as tk
from tkinter import filedialog
from PIL import Image, ImageTk
import numpy as np

window = tk.Tk()
window.title("Vision Based Garbage Segregation")
window.geometry("650x750")

def classify_image():

    file_path = filedialog.askopenfilename(
        title="Select Garbage Image",
        filetypes=[
            ("Image Files", "*.jpg *.jpeg *.png")
        ]
    )

    if not file_path:
        return

    # Open image
    img = Image.open(file_path).convert("RGB")

    # Display image
    display_img = img.copy()
    display_img.thumbnail((450, 450))

    photo = ImageTk.PhotoImage(display_img)

    image_label.config(image=photo)
    image_label.image = photo

    # Resize image
    small_img = img.resize((100, 100))

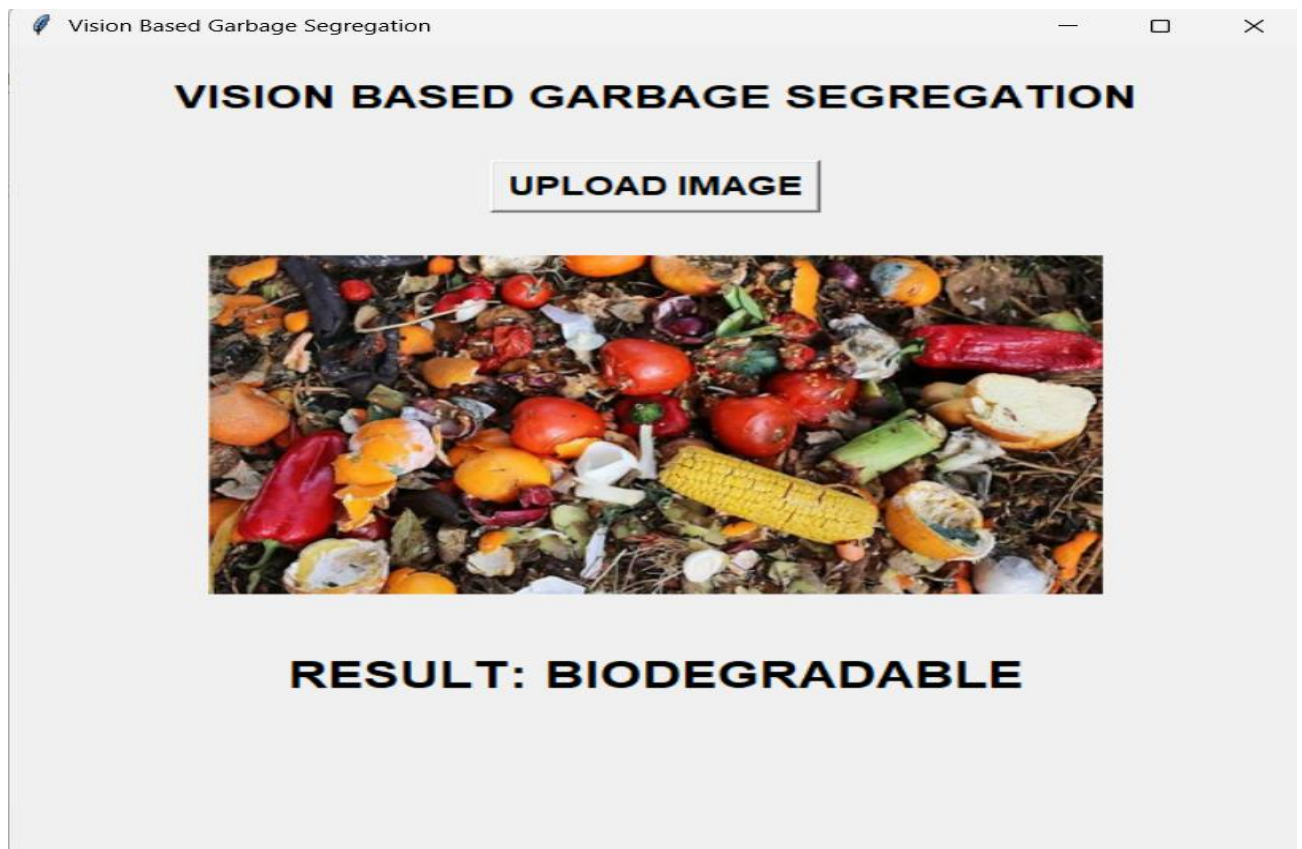
    # Convert to NumPy array
    pixels = np.array(small_img)
```

Ln: 1 Col: 0

4.3 Test Plan & Verification

Requirement		Test Method		Acceptance Criterion	
Image upload		Upload valid JPG/PNG		Image accepted	
Specification	Required Value		Achieved Value		Status (Pass/Fail)
Supported image upload					
Image validation					

4.4 Results



4.5 Data Analysis & Engineering Judgment

The preprocessing module provides a standardized interface between raw user images and the future garbage classification model. The primary engineering benefit is reduction of variation in image dimensions and numerical representation. Without preprocessing, images from different sources can have different dimensions and pixel characteristics. A fixed preprocessing pipeline ensures that the subsequent classification stage receives data in a predictable format. However, preprocessing also involves trade-offs. Excessive resizing can remove important visual details, while aggressive normalization or transformation may change characteristics relevant to classification. Therefore, preprocessing parameters should ultimately be selected based on the requirements and validation performance of the final classification model.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design image upload interface	Working upload mechanism	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Easy image upload.
- Improves image quality.
- Removes image noise.
- Resizes images for uniform processing.
- Enhances important image features.
- Prepares images for accurate classification.

Limitations

- Requires a clear image.
- Poor lighting can affect results.
- Blurred images may reduce accuracy.
- Large images may take more processing time.
- Different image formats may need conversion.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
Student 1	Complete image-uploading and preprocessing module, testing and documentation	100%
Item	Estimated Cost	Actual Cost
Laptop/Computer	Existing	Existing

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project developed an Image Uploading and Preprocessing Module as the initial stage of a Vision-Based Smart Garbage Segregation System. The module provides a structured process for accepting waste images, validating the input and converting images into a standardized representation for subsequent computer-vision processing. The implementation includes image uploading, validation, image loading, resizing, colour processing and normalization. These operations establish a consistent input format and provide a suitable interface for future machine-learning-based garbage classification. The project demonstrates that a modular preprocessing pipeline can simplify the integration of a future waste-classification model while improving consistency between raw images and model-ready inputs. The current work is limited to image preprocessing and therefore does not claim complete automated waste classification or physical segregation.

5.2 Future Work

The project can be extended by:

- Integrating a CNN-based waste classification model.
- Training the model using a suitable waste-image dataset.
- Classifying waste into biodegradable, recyclable and non-recyclable categories.
- Adding confidence scores to predictions.
- Implementing real-time camera input.
- Adding object detection for images containing multiple waste objects.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

1. Image-processing fundamentals.
2. Image uploading and validation.
3. Image resizing.
4. Colour-space processing.

5. Pixel normalization.

Engineering & Professional Skills Developed:

1. Requirements analysis.
2. Modular system design.
3. Software implementation.
4. Testing and verification.

Individual Reflection (each student, 4–5 lines):

The main engineering challenge was designing a preprocessing pipeline that could handle images with different formats, resolutions and dimensions while producing a consistent output. This was addressed by introducing validation and standardized image-processing stages before the image was passed to the next stage.

A key engineering decision was to separate image uploading from preprocessing rather than combining all operations into one function. This improves modularity and makes it easier to replace or modify individual preprocessing operations.

During the project, I independently learned how digital images are represented as numerical arrays and how resizing, colour conversion and normalization affect the data supplied to machine-learning models.

If the project were redesigned, I would add automated image-quality checking and evaluate different preprocessing techniques quantitatively using the performance of the final waste-classification model.

REFERENCES

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- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in **Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)**, 2016, pp. 770–778.
- [4] F. Thung and M. Yang, “Classification of trash for recyclability status,” Stanford University, CS229 Project Report, 2016.
- [5] OpenCV Documentation, “Open Source Computer Vision Library,” documentation for image processing and computer vision functions.
- [6] Python Software Foundation, “Python Documentation,” Python programming language documentation.
- [7] NumPy Developers, “NumPy Documentation,” documentation for numerical computing and array processing.
- [8] ISO/IEC 25010, **Systems and Software Engineering—Systems and Software Quality Requirements and Evaluation (SQuaRE)—Product Quality Model**.
- [9] ISO/IEC/IEEE 29148, **Systems and Software Engineering—Life Cycle Processes—Requirements Engineering**.
- [10] The dataset documentation corresponding to the waste-image dataset actually used in the project, [Dataset Name], [Dataset Source], [Year].

APPENDICES

Appendix A – Detailed Calculations

For an 8-bit image:

$$I_{normalized} = \frac{I_{original}}{255}$$

For example, if a pixel value is:

$$I = 128$$

then:

$$I_{normalized} = \frac{128}{255} = 0.502$$

Therefore, the pixel is represented approximately as **0.502** after normalization.

Appendix B – Engineering Drawings / Schematics

Appendix C – Source Code / Code Repository Information

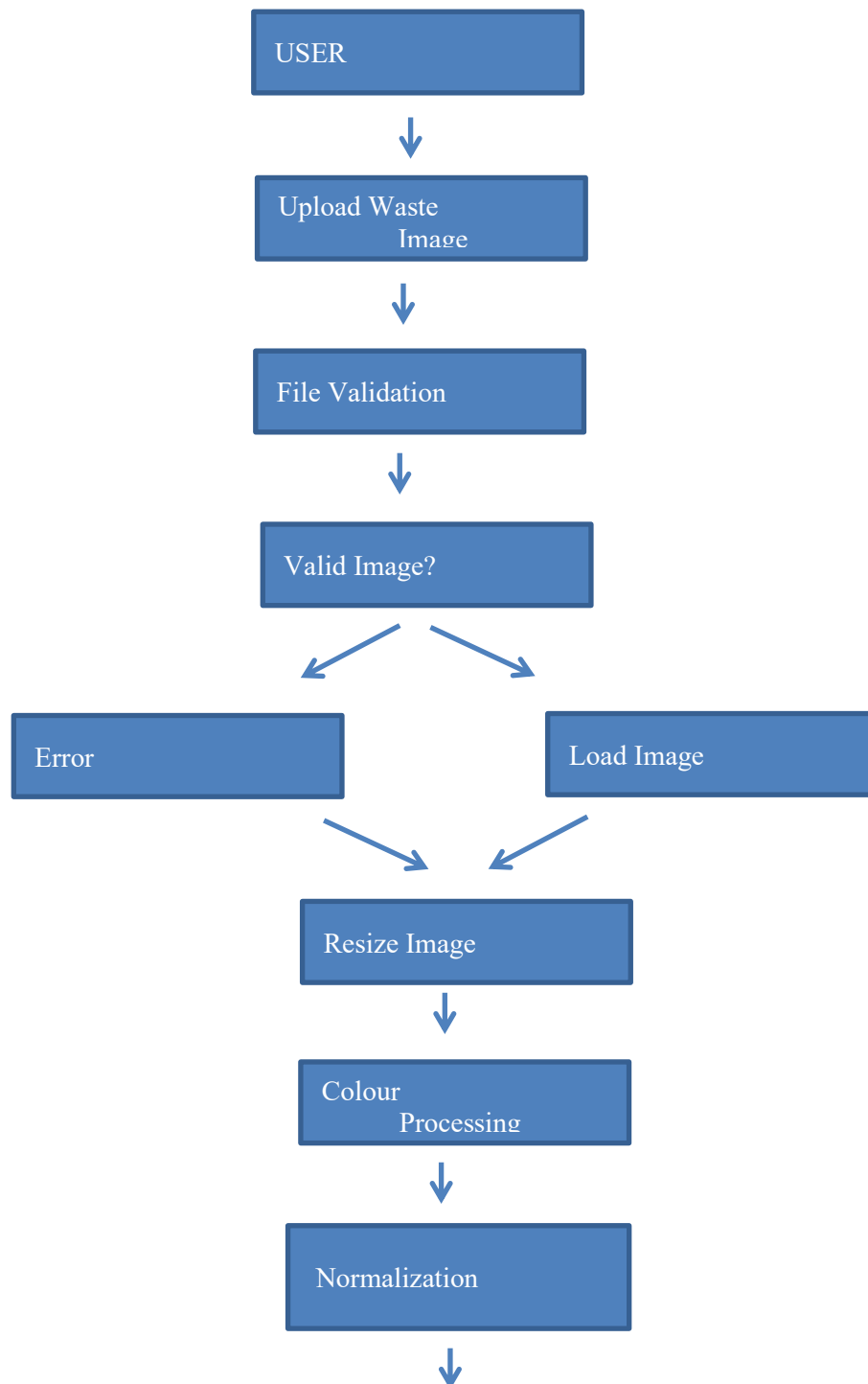
SmartGarbageSegregation/

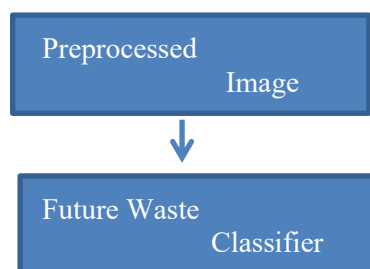
```
|
|— app.py
|— preprocessing.py
|— requirements.txt
|
|— static/
|   |— style.css
|
|— templates/
|   |— index.html
|
|— uploads/
|
|— processed/
```

Appendix D – Datasheets

1. Dataset name
2. Source
3. Number of images
4. Waste categories
5. Image formats
6. Training/testing split
7. License

Appendix E – Experimental / Raw Data





Appendix F – Ethics / Institutional Approval

The current project is a software-based image-processing system and does not involve human clinical experimentation or direct handling of hazardous waste during testing. Uploaded images should be handled responsibly, and unnecessary personal or identifying information contained in images should not be stored.

Appendix G – Project Meeting / Progress Records

Date	Activity	Outcome
[Date]	Problem identification	Project topic finalized
[Date]	Literature review	Existing solutions reviewed
[Date]	Requirement analysis	Module requirements defined
[Date]	Upload interface	Upload functionality developed
[Date]	Preprocessing	Image processing implemented
[Date]	Testing	Test cases performed
[Date]	Debugging	Errors corrected
[Date]	Documentation	Report prepared

Appendix H – User/Industry Feedback

User	Feedback	Action Taken
User 1	Upload interface should be simple	Interface simplified
User 2	Processing output should be visible	Output preview added

Appendix I – Publications / Patents / Competitions / Product Outcomes

No publication, patent, competition award or commercial product outcome was associated with the current implementation at the time of report submission.

Appendix J – Individual Contribution Evidence

Student 1 contributed to the Vision-Based Smart Garbage Segregation System by developing the image uploading and preprocessing module. The contribution included designing the image input workflow, implementing image upload and validation for supported formats such as JPG and PNG, resizing images to the required dimensions, performing colour processing and normalizing pixel values for further analysis. PAVAN JANA also conducted functional testing using different waste images, identified and corrected image-processing errors, and verified the preprocessing outputs. In addition, the student contributed to the system design, testing and analysis sections of the project report and supported the integration of the preprocessing module with the subsequent garbage classification stage.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)